SUPPLIER HEATMAP | SUPPLIER 1 TO BE ADRESSED IN PRIORITY

PRELIMINARY

Risk concentration

- % parts > 20%
- % parts > 10%
- % parts < 10%

OTD

- OTD < 60%
- OTD > 60%
- OTD > 85%

Average

- Monthly Avg. < 8
- Monthly Avg. > 8
- Monthly Avg. > 10

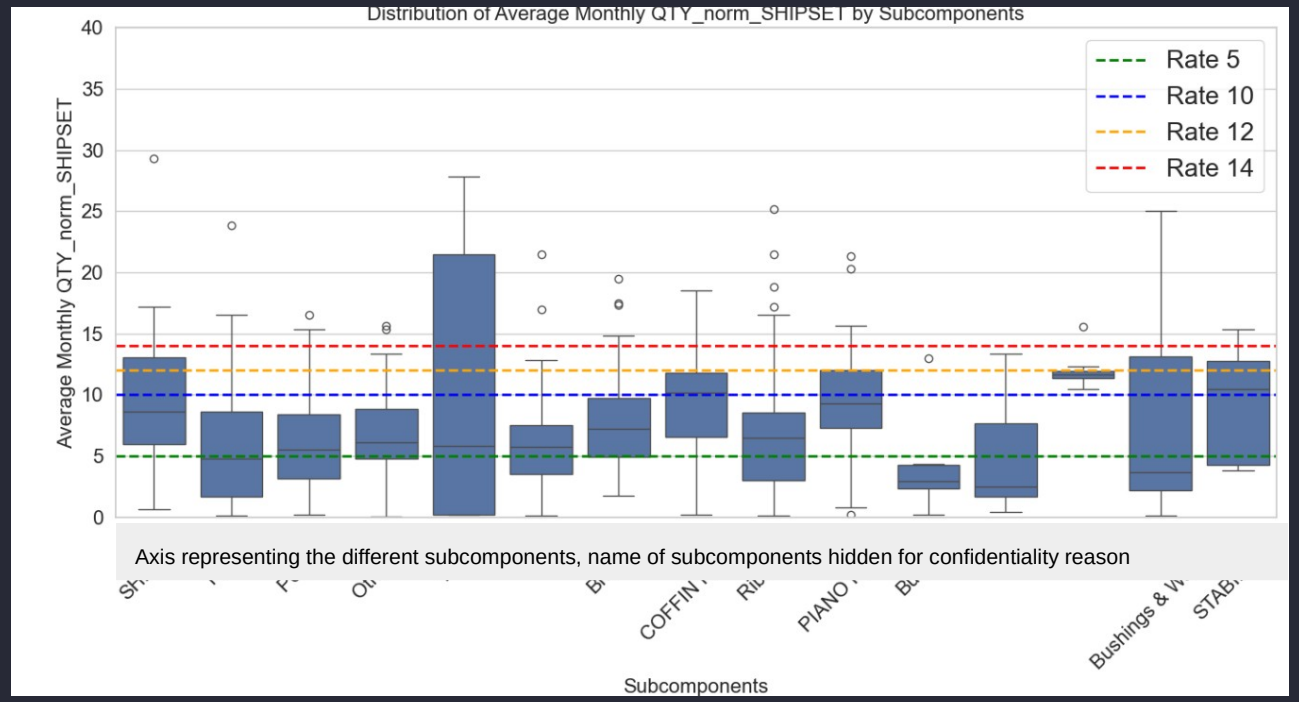
Supplier	Risk concentration	OTD	Average rate	SYNTHESIS
SUPP 1				
SUPP 2				
SUPP 3				
SUPP 4				
SUPP 5				
SUPP 6				
SUPP 7				

PRELIMINARY VERSION

RATE 12 ENABLEMENT | PARTS RECEIVED – RATE CALCULATION - SUBCOMPONENTS

Component 1 + Component 2

Reception – Average[1] rate volume based (monthly) of parts received between 02/25 and 07/25 – Grouped by subcomponents



Parameters of the distribution

Subcomponent	Q1 (25 Pctl)	Median	Q3 (75 Pctl)
Sub 1	4,96	7,22	9,71
Sub 2	5,98	8,58	13,08
Sub 3	1,67	4,75	8,63
Sub 4	2,98	6,50	8,54
Sub 5	3,13	5,50	8,42
Sub 6	4,79	6,08	8,88
Sub 7	3,54	5,71	7,50
Sub 8	1,67	2,50	7,67
Sub 9	7,32	9,29	12,04
Sub 10	6,58	10,17	11,83
Sub 11	2,15	3,66	13,10
Sub 12	0,17	5,83	21,50
Sub 13	11,33	11,67	11,92
Sub 14	4,25	10,50	12,75
Sub 15	2,33	2,92	4,25

Key insights

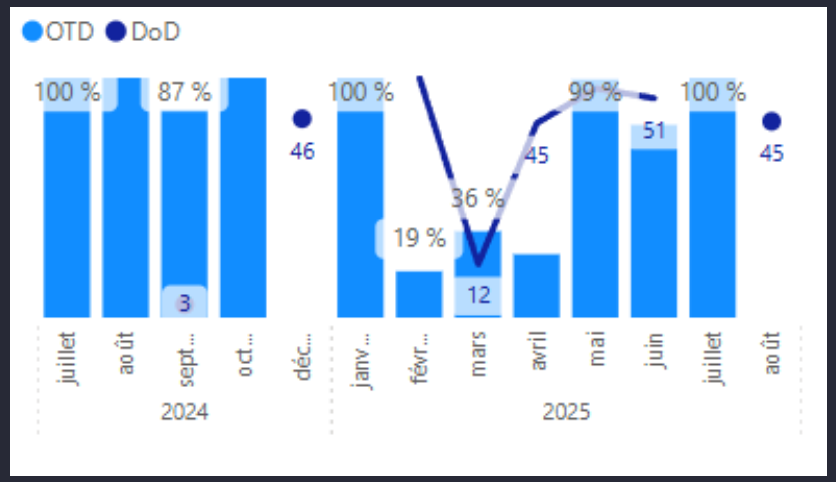
- For all the subcomponents to be enabled at **Rate 12, Q1 should be close to the median**, which in turn should be close to the targeted rate
- Although the subcomponents on Component 2 are performing better, with the **median above 8**, **Q1 is still too low**) for full shipset enablement
- The **Component 1** are struggling even more and need to be investigated

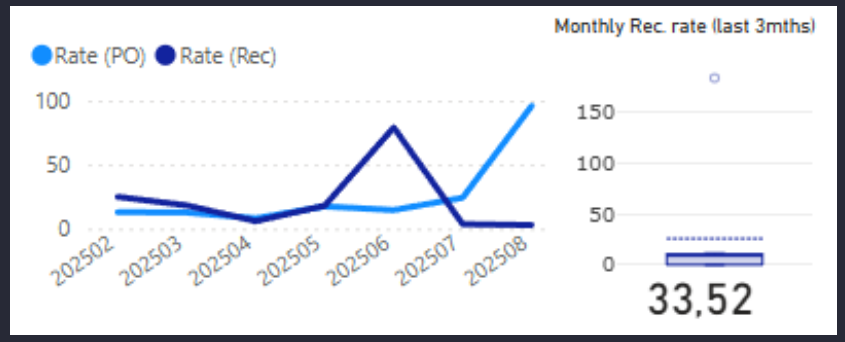
1. Volume based calculation method

PRELIMINARY VERSION

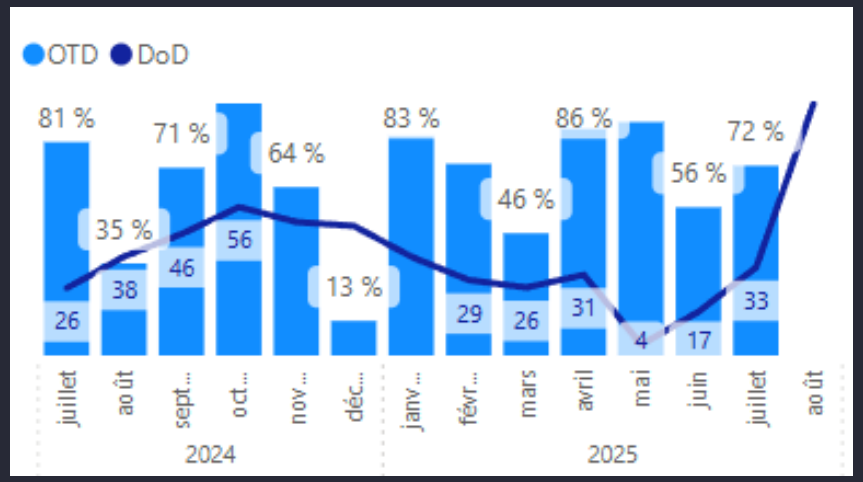
SUPPLIER DETAILED ANALYSIS | SUPPLIER 3 (EVOLUTION)

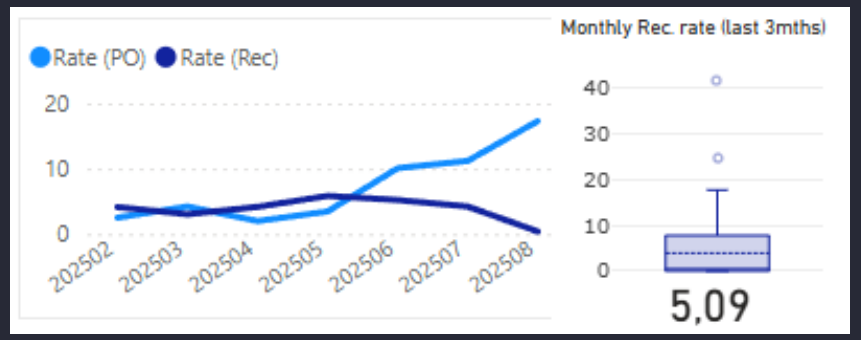
Loose items





FTE





PRELIMINARY VERSION

An end-to-end framework is needed to analyze suppliers' capacity to achieve Rate 14

Supplier category analysis	Critical capacity factors
Manufacturing parts suppliers	• Machine **availability** and cycle times • **Skilled labor** availability • **Raw material** supply chain stability • **Quality** control processes • **Tooling** and fixture capacity
Surface treatment suppliers	• **Chemical processing** line capacity • Cure/drying cycle times • **Skilled labor** availability • **Batch processing** line capacity • **Environmental** compliance capacity **(?)** • **Quality testing** throughput • **Specialized** equipment availability (e.g. anodizing)

W1 W2 W3

Demonstrated capacity assessment on the **top 3 suppliers** → Estimated capacity assessment on **the top 3 suppliers**

Risk assessment on the top 3 suppliers

Demonstrated capacity metrics
- **Peak Monthly Output**: Highest monthly production achieved
- **Sustained Output**: Consistent monthly production over 6+ consecutive months
- **Quality Yield Rate**: % of units passing quality standards at peak production manufacture
- **On-Time Delivery Rate**: % of deliveries meeting schedule at peak volumes
- **Resource Utilization**: Labor, equipment, and facility utilization at peak

Estimated capacity metrics
- **Theoretical Capacity**: Maximum output based on equipment specifications
- **Effective Capacity**: Realistic maximum considering efficiency factors *(Theoretical x Equipment Efficiency x Human Efficiency x Quality Yield)*
- **Investment-Enhanced Capacity**: Projected capacity after planned improvements

Risk assessment
- **Technical risks**: Process scalability, equipment reliability, quality consistency
- **Operational risks**: Supply chain disruption, labor shortage, ability to manage modification
- **Financial risks**: Financial health (e.g., cash flow)
- **Strategic risks**: Dependency risk (e.g., over-reliance on single suppliers)

- For some suppliers, an interview-based **process mapping** will be need to prioritize some critical capacity factors
- Outputs could be presented in form of **heatmap** with 3 colors (green, amber, red) with data driven justification

AGENDA

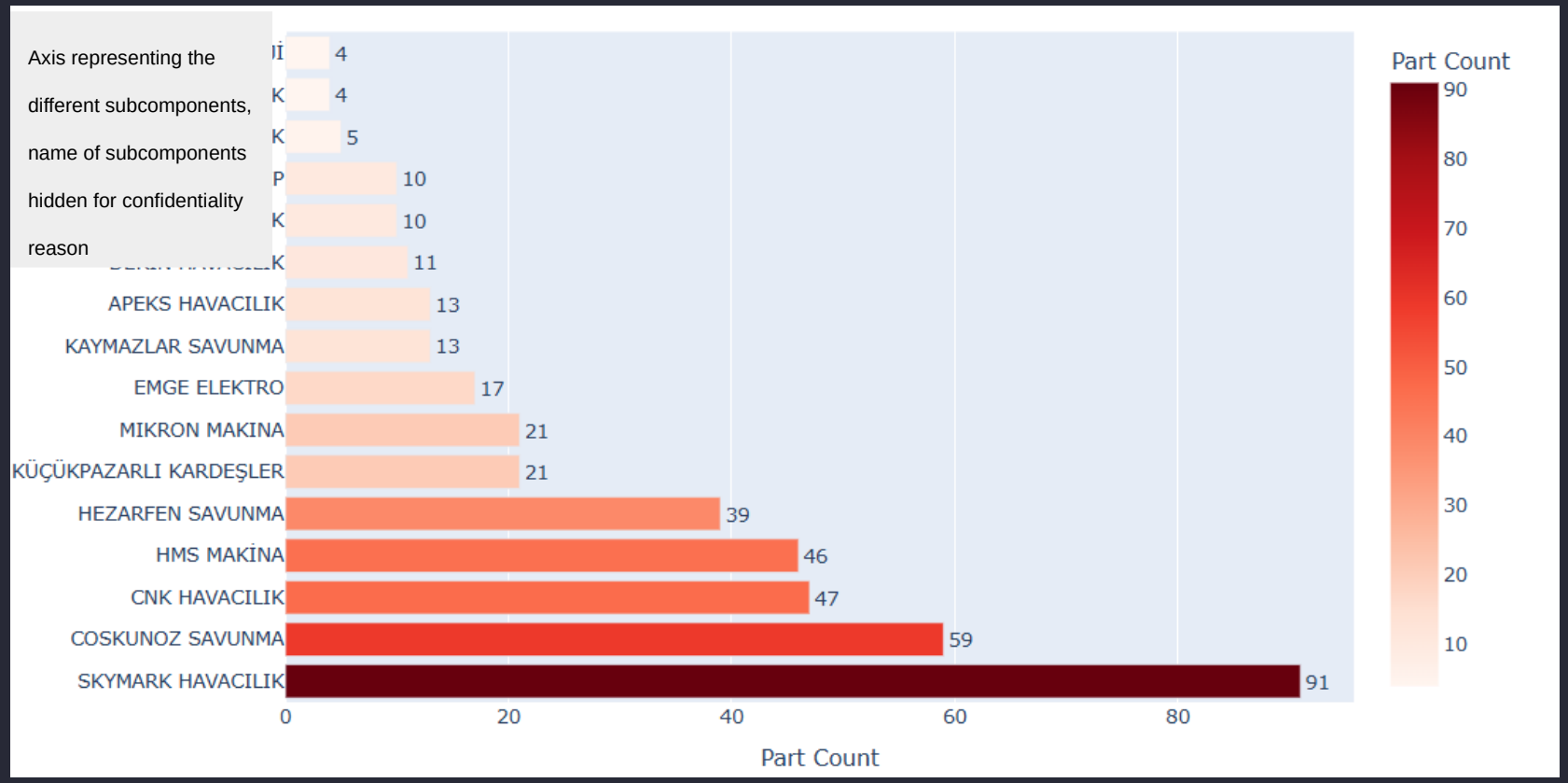
1. Context
2. Rate Analysis
3. Data inconsistencies analysis
4. Capacity Management Analysis (CMA)
5. Missing Part Analysis (1st Part)
6. **Missing Part Analysis (2nd Part)**
7. Pain points with TAI
8. Pain points with TAI suppliers
9. Template of Supply Chain Report

PRELIMINARY VERSION

MISSING PART ANALYSIS | PARETO OF SUPPLIERS

FTE

Pareto of Suppliers on External SC – MSN 400 to 428



Key insights

- Supplier 1 alone represents **22% of all the missing parts**
- Suppliers 1, 2, 3 represent **60% of all the missing parts**

PRELIMINARY VERSION